

WISP

Wireless Intelligent Sensing for Personalized Self-Triage

Progress report: what the system looks like today and what has been built so far.

When “I just feel weak” is all you can say, WISP can check whether your physical function is actually different from your usual self, then tell you what to do next and why.

Prototype status as of 4 October 2026 · Screens captured from the running system (web app + local API, recorded sensor replay).

The sensor recordings used in these screenshots are **synthetic** and are labelled as such in the UI.

1. Summary

WISP is a self-triage and care-navigation agent for older adults in Singapore with vague complaints such as “I feel weak” or “I don’t feel like myself”. It asks one question at a time, screens for emergency warning signs with fixed rules, and then decides whether a short physical check — the **Five-Times Sit-to-Stand (5xSTS)**, measured contactlessly with Wi-Fi — would actually change the advice. If it would, and it is safe, WISP runs the check, compares it with the person’s own usual result, and gives one of five care decisions with what to do, when, and why.

The central idea is visible in the product: **the AI sometimes chooses to measure and sometimes deliberately chooses not to**. A physical result can raise urgency but can never lower it below what the symptoms already require.

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automated backend tests passing
(safety rules, ordering, security,
sensing, scenarios, API)

4 / 4

demo scenarios run end-to-end
through the real browser UI

15 / 15

test conversations reached the right
care level and the right measure /
don't-measure choice

Five possible outcomes

TIER	PATIENT SEES	SINGAPORE ACTION
T1 Emergency	This needs help now	Call 995 or go to A&E
T2 Same-day	Please be seen today	GP or polyclinic today; A&E if unavailable
T3 Primary care soon	Book your doctor in the next few days	Regular GP / polyclinic within 2–3 days
T4 Self-care	You can continue monitoring at home for now	Home care, warning signs, re-check tomorrow 10 AM
ABSTAIN	I can't safely judge this from here	Speak to family doctor, nurse or caregiver today

How a check flows

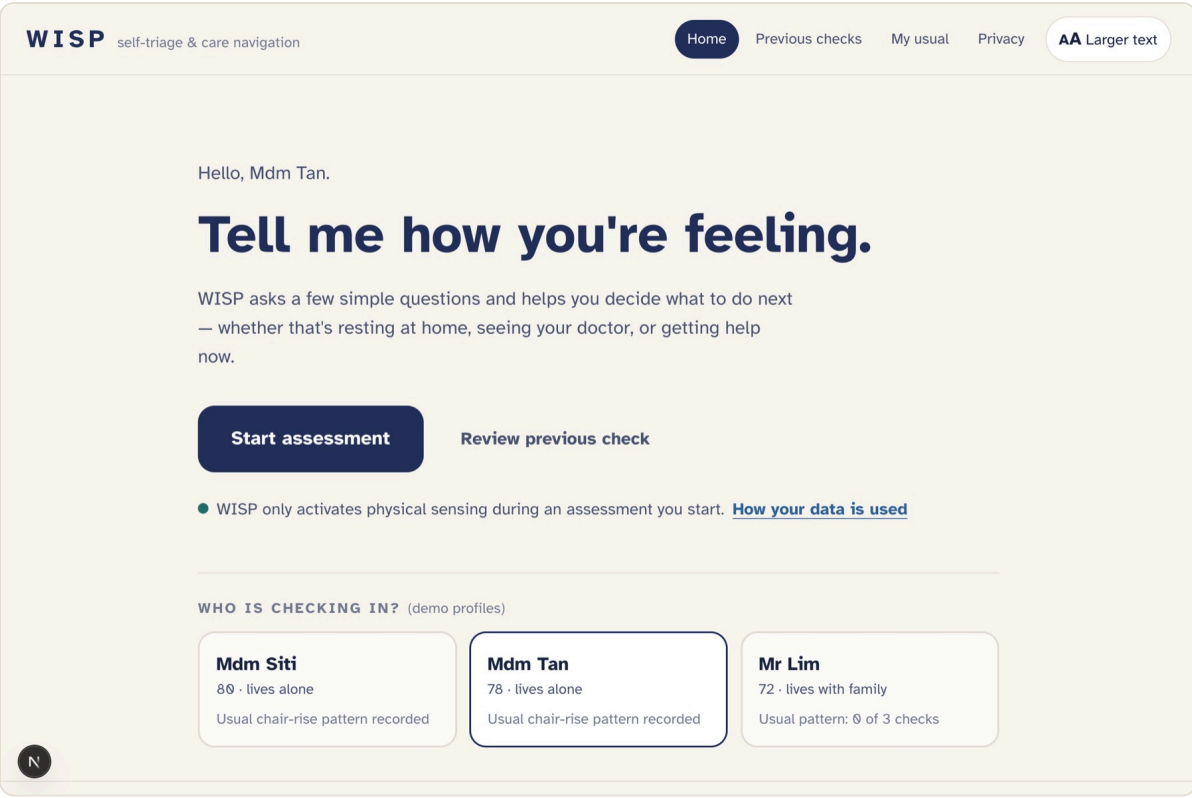
Patient complaint → one-question-at-a-time assessment → deterministic red-flag screen

- possible care range (floor & ceiling) → would a physical check change the answer?
- if yes and safe: 5xSTS via Wi-Fi sensing → compare with personal baseline
- deterministic care-tier engine → explanation, care navigation, optional re-check

2. What has been developed so far

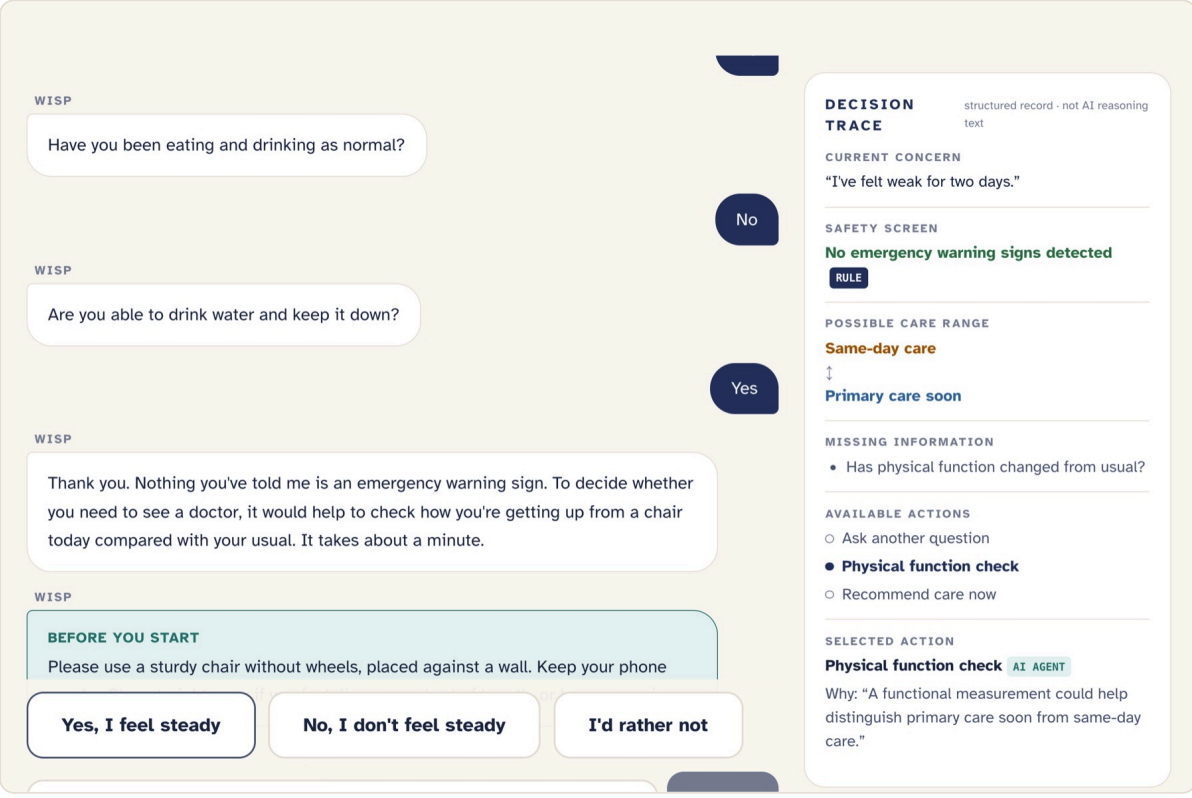
AREA	WHAT EXISTS	STATUS
Patient web app	Next.js app: welcome, one-question chat with large answer buttons, physical check screen, recommendation screen, decision trace, previous checks, baseline enrolment, privacy & data controls, caregiver summary, larger-text mode, phone layout.	Done
Safety rules engine	Red-flag screen (9 warning signs), symptom care range, eligibility gate, care-tier engine, abstention. Asymmetric rule: sensing never lowers urgency.	Done
Agent tools (MCP)	13 tools for Tencent WorkBuddy over MCP (stdio and HTTP), with ordering enforced server-side: one-time permission to sense, signed measurements, sensor lock after any warning sign.	Done
Built-in agent	Offline stand-in that runs the conversation and calls the same tools, so the demo works without network. Optional LLM extraction (e.g. Hunyuan) can add, never remove, warning signs.	Done
WorkBuddy connection	MCP server, agent instructions (skill) and config written; verified with a scripted MCP client. Not yet connected to WorkBuddy itself.	Needs live test
Wi-Fi sensing pipeline	Signal processing on CSI amplitude (filtering, PCA, motion energy, posture signal, five-rise detection, timing, confidence, second-person checks). Sensor-independent provider interface.	Done
ESP32 live capture	Serial reader for ESP32 CSI output, live provider, raw-capture saving for debugging. Parser tested; no hardware run yet.	Needs hardware
Personal baseline	Enrolment over 3 well-day checks, encrypted storage, comparison labels (within / mildly / clearly slower, unable to compare, unreliable).	Done
Re-assessment	T4 schedules a re-check; follow-up check shows yesterday as context only; new warning signs override a previous normal result.	Done
Privacy	Raw sensing stays local; agent receives summaries only; consent per caregiver share; delete-my-data; visible sensing state OFF / ACTIVE / COMPLETE / LOCKED.	Done
Developer / demo tools	Sensor source and replay controls, “someone walks through” demo, signal charts, measurement table, audit log, ground-truth entry for accuracy.	Done
Evaluation	Test conversations, fairness variants, warning-sign extraction, sensor accuracy (synthetic + real-participant template).	Real data pending
Documentation	README, architecture, safety rules, privacy, demo script, WorkBuddy setup, ESP32 setup; one-command start script.	Done

3. The system today



The WISP home screen features a header with the logo 'WISP self-triage & care navigation' and navigation links: Home, Previous checks, My usual, Privacy, and a text size selector 'AA Larger text'. The main content area greets the user 'Hello, Mdm Tan.' and displays a large heading 'Tell me how you're feeling.' Below this, a subtext explains that WISP asks simple questions to help decide the next steps. Two primary buttons are present: 'Start assessment' and 'Review previous check'. A privacy note states: 'WISP only activates physical sensing during an assessment you start. [How your data is used](#)'. A section titled 'WHO IS CHECKING IN? (demo profiles)' shows three user cards: Mdm Siti (80, lives alone, usual chair-rise pattern recorded), Mdm Tan (78, lives alone, usual chair-rise pattern recorded), and Mr Lim (72, lives with family, usual pattern: 0 of 3 checks). A small 'N' icon is in the bottom left corner.

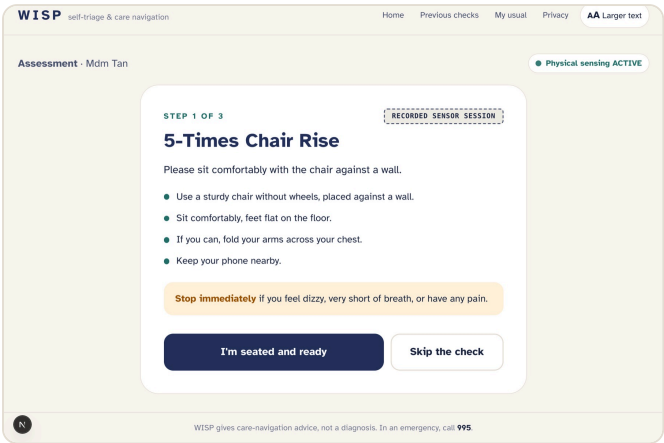
Welcome. Leads with “Tell me how you’re feeling”, not with sensing. One primary action, a privacy note, and demo profiles (Mdm Tan, Mr Lim, Mdm Siti).



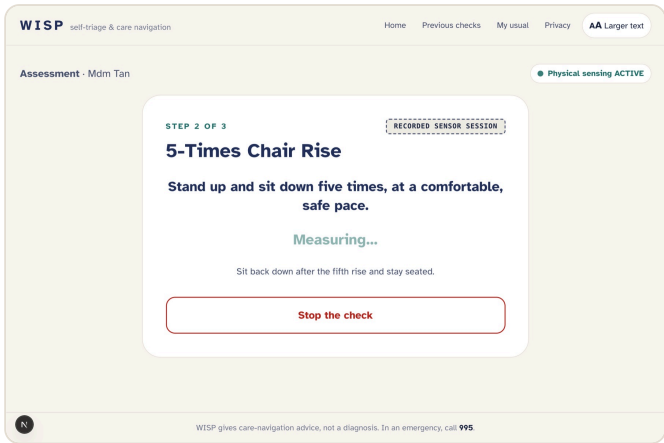
The assessment interface shows a series of questions from WISP: 'Have you been eating and drinking as normal?' (with 'No' and 'Yes' buttons), 'Are you able to drink water and keep it down?' (with 'No' and 'Yes' buttons), and a longer question about getting up from a chair. A 'BEFORE YOU START' section instructs the user to use a sturdy chair. At the bottom are three large buttons: 'Yes, I feel steady', 'No, I don't feel steady', and 'I'd rather not'. On the right, a 'DECISION TRACE' panel provides a structured record of the reasoning. It includes the current concern ('I've felt weak for two days.'), a passed safety screen, a possible care range from 'Same-day care' to 'Primary care soon', missing information (physical function change), and the selected action 'Physical function check' by the AI agent, with a brief explanation of why.

Assessment with Decision Trace. One question at a time with large answer buttons. The right panel shows the structured decision record: safety screen passed, possible care range, missing information, and the selected action — *Physical function check* — with the reason.

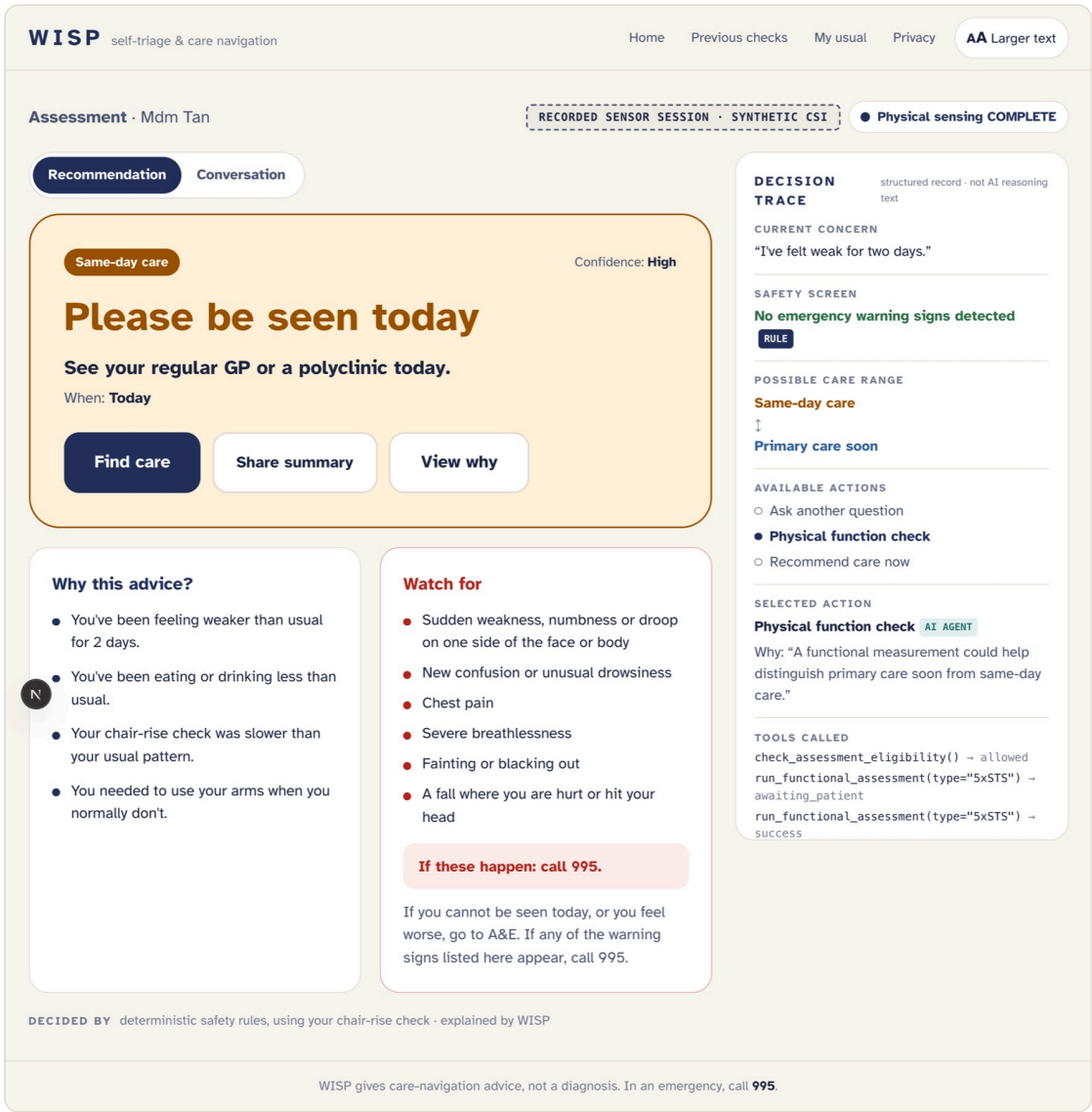
Scenario 1 — the agent chooses to measure (Mdm Tan, 78: “I’ve felt weak for two days”, eating less)



Physical check, step 1. Chair against a wall, stop if unwell. The patient starts it themselves. “Recorded sensor session” is always shown on replay.



Physical check, step 2. Calm “Measuring...” ; no waveforms shown to the patient. Sensing indicator shows ACTIVE.



Recommendation: Please be seen today (T2). The action is the centre of the screen, not the measurement. Reasons in plain language, warning signs, and the trace showing the decision impact Primary care soon → Same-day care.

Scenario 2 — the agent refuses to measure (Mr Lim, 72: sudden dizziness, clumsy left hand)

WISPself-triage & care navigation

HomePrevious checksMy usualPrivacyAA Larger text

Assessment · Mr Lim

Physical sensing LOCKED for this check

RecommendationConversation

Emergency

Confidence: High

This needs help now

Call 995 now, or go to the nearest A&E.

When: Now

Call 995

Show nearest emergency department

View why

Demo: WISP never places a call by itself. Tapping "Call 995" opens your phone dialler.

N

Why this advice?

This needs urgent medical attention now because your symptoms started suddenly and you have weakness, numbness or clumsiness on one side.

While you wait for help

Do not drive yourself. If you are alone, call 995 first, then unlock your door if you can.

DECIDED BYdeterministic safety rules · explained by WISP

WISP gives care-navigation advice, not a diagnosis. In an emergency, call 995.

DECISION TRACEstructured record · not AI reasoning text

CURRENT CONCERN
"This morning I suddenly felt dizzy and my left hand feels clumsy."

SAFETY SCREEN
Emergency warning sign: Sudden onset, One-sided weaknessRULE

POSSIBLE CARE RANGE
Emergency

AVAILABLE ACTIONS

Ask another question

Physical-function-check

Recommend care now

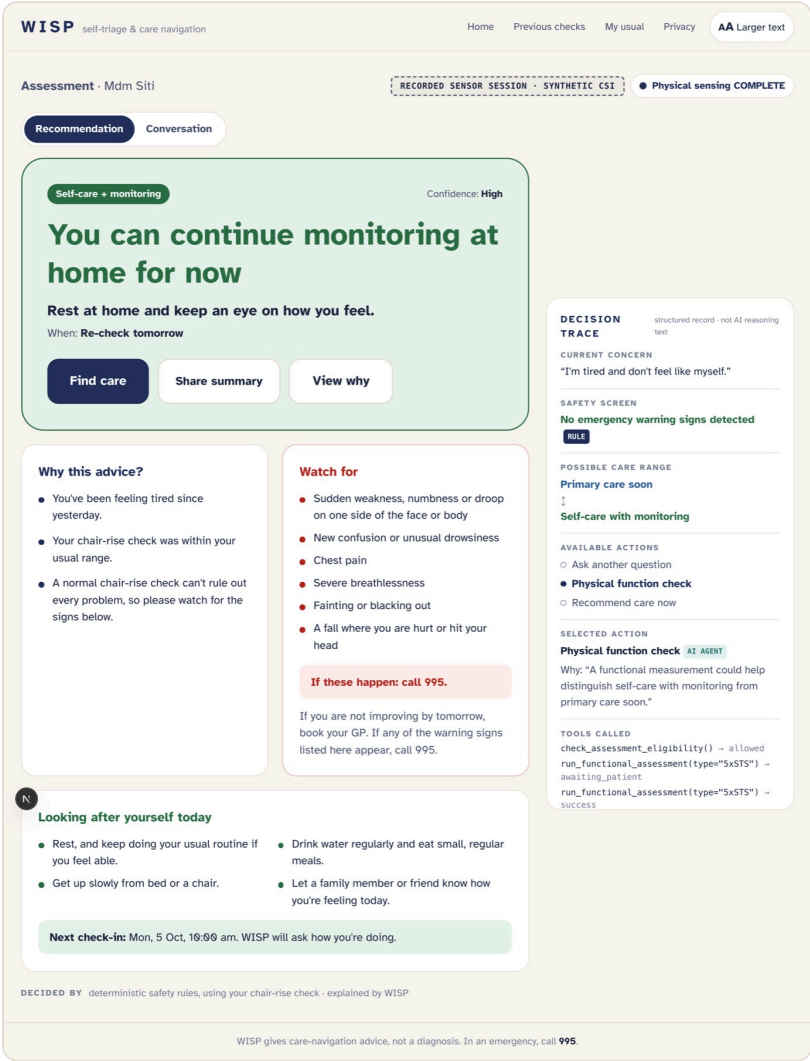
SENSING NOT REQUESTED
Emergency warning sign already determines the care recommendation.

SELECTED ACTION
Escalate nowAI AGENT
Why: "Emergency warning sign reported. Physical sensing not requested: it cannot change an emergency recommendation."

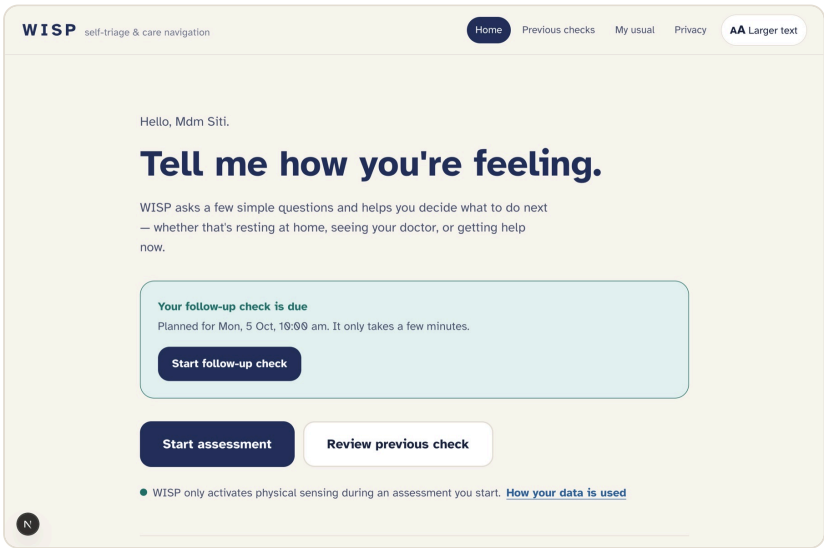
TOOLS CALLED
decide_care_tier() — T1

Emergency (T1). Call 995 and nearest emergency department. The trace shows **SENSING NOT REQUESTED** and the sensor is LOCKED: an emergency warning sign already decides the advice. WISP never dials automatically.

Scenario 3 — re-assessment (Mdm Siti, 80: “I’m tired and don’t feel like myself”)



Day 1: self-care with monitoring (T4). Chair-rise within her usual range. Self-care advice, warning signs, and a re-check booked for tomorrow 10 AM.



Follow-up due. The home screen invites the scheduled re-check.

WISP

self-triage & care navigation

[Home](#)
[Previous checks](#)
[My usual](#)
[Privacy](#)

AA Larger text

Follow-up check · Mdm Siti

Recommendation

Conversation

Emergency

Confidence: High

This needs help now

Call 995 now, or go to the nearest A&E.

When: Now

Call 995

Show nearest emergency department

View why

Demo: WISP never places a call by itself. Tapping "Call 995" opens your phone dialler.

Why this advice?

This needs urgent medical attention now because there is new confusion or unusual drowsiness.

While you wait for help

Do not drive yourself. If you are alone, call 995 first, then unlock your door if you can.

DECIDED BY

deterministic safety rules · explained by WISP

Physical sensing LOCKED for this check

DECISION TRACE

structured record · not AI reasoning text

PREVIOUS CHECK

Sun, 4 Oct: You can continue monitoring at home for now · chair-rise: within your usual range

A previous result is context only. It is never used to lower today's urgency.

CURRENT CONCERN

"My daughter said I seemed confused last night."

SAFETY SCREEN

Emergency warning sign: Confusion

RULE

POSSIBLE CARE RANGE

Emergency

AVAILABLE ACTIONS

Ask another question

Physical-function-check

Recommend care now

SENSING NOT REQUESTED

Emergency warning sign already determines the care recommendation.

SELECTED ACTION

Escalate now

AT AGENT

WISP gives care-navigation advice, not a diagnosis. In an emergency, call 995.

Day 2: “My daughter said I seemed confused last night” → Emergency (T1). The trace shows yesterday’s normal result as context only — it is never used to lower today’s urgency.

WISP — progress report

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Scenario 4 — someone walks through during the check

WISPself-triage & care navigation

HomePrevious checksMy usualPrivacyAA Larger text

Assessment · Mdm Tan

RECORDED SENSOR SESSION · SYNTHETIC CSI

Physical sensing COMPLETE

RecommendationConversation

Cannot safely assess

Confidence: Low

I can't safely judge this from here

Please speak with your family doctor or a nurse today about how you're feeling.

When: Today

Find careShare summaryView why

Why this advice?

N

- You've been feeling weaker than usual since yesterday.
- I couldn't get a reliable chair-rise reading, so I didn't use it.

Watch for

- Sudden weakness, numbness or droop on one side of the face or body
- New confusion or unusual drowsiness
- Chest pain
- Severe breathlessness
- Fainting or blacking out
- A fall where you are hurt or hit your head

If these happen: call 995.

If any of the warning signs listed here appear, call 995.

DECIDED BYdeterministic safety rules - explained by WISP

WISP gives care-navigation advice, not a diagnosis. In an emergency, call 995.

DECISION TRACE

structured record - not AI reasoning text

CURRENT CONCERN

"I feel weak."

SAFETY SCREEN

No emergency warning signs detected

RULE

POSSIBLE CARE RANGE

Primary care soon

Self-care with monitoring

AVAILABLE ACTIONS

Ask another question

Physical function check

Recommend care now

SELECTED ACTION

Physical function checkAI AGENT

Why: "A functional measurement could help distinguish self-care with monitoring from primary care soon."

TOOLS CALLED

check_assessment_eligibility() → allowed

run_functional_assessment(type="5xSTS") → awaiting_patient

run_functional_assessment(type="5xSTS") → multiple_people_detected

Measurement rejected → “I can’t safely judge this from here”. The second moving person was detected (multiple_people_detected), the reading was not used, and WISP advises speaking to a doctor today rather than defaulting to home care.

No, I'm alone

WISP

Thank you. Your screen will now show the steps for the chair-rise check.

WISP

I couldn't get a clear reading, so I won't use that result. I'll base my advice on what you've told me.

WISP

I want to be honest with you: i can't safely judge this from here. Please speak with your family doctor or a nurse today about how you're feeling.

WISP

Would you like to share a short summary with Daniel?

Yes, share itNo, thank you

Or type your answer...

Send

N

call 995

DECISION TRACE

structured record - not AI reasoning text

CURRENT CONCERN

"I feel weak."

SAFETY SCREEN

No emergency warning signs detected

RULE

POSSIBLE CARE RANGE

Primary care soon

Self-care with monitoring

AVAILABLE ACTIONS

Ask another question

Physical function check

Recommend care now

SELECTED ACTION

Physical function checkAI AGENT

Why: "A functional measurement could help distinguish self-care with monitoring from primary care soon."

TOOLS CALLED

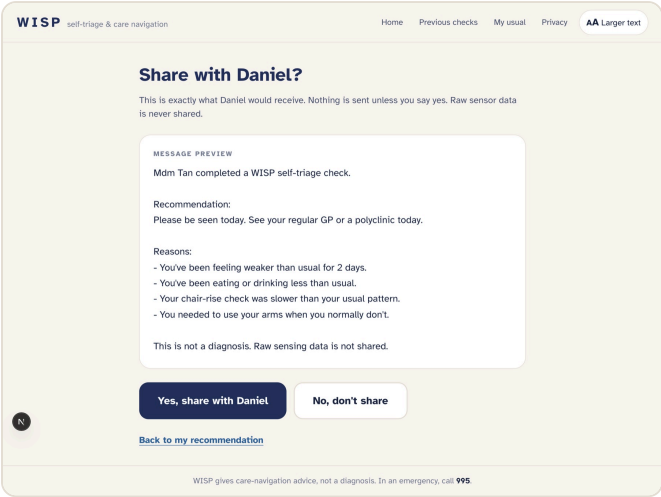
check_assessment_eligibility() → allowed

run_functional_assessment(type="5xSTS") → awaiting_patient

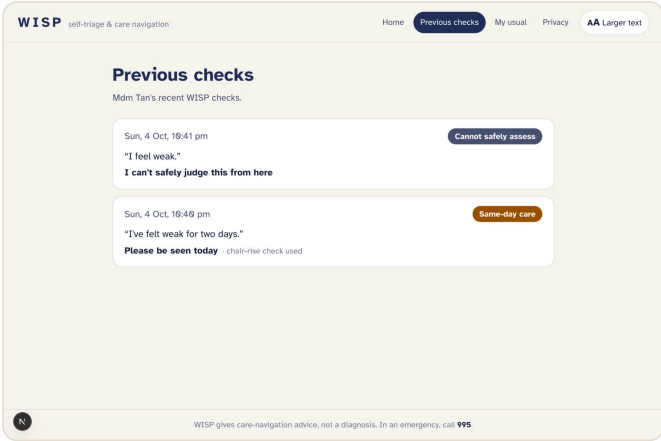
run_functional_assessment(type="5xSTS") → multiple_people_detected

Conversation view. “I couldn’t get a clear reading, so I won’t use that result.”

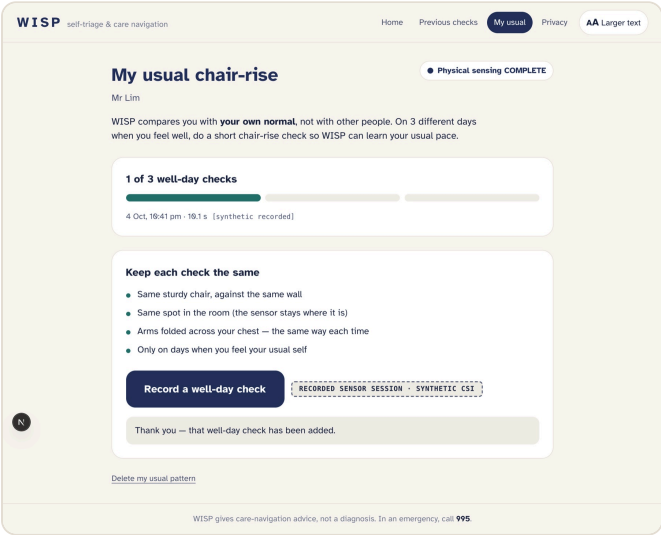
Supporting screens



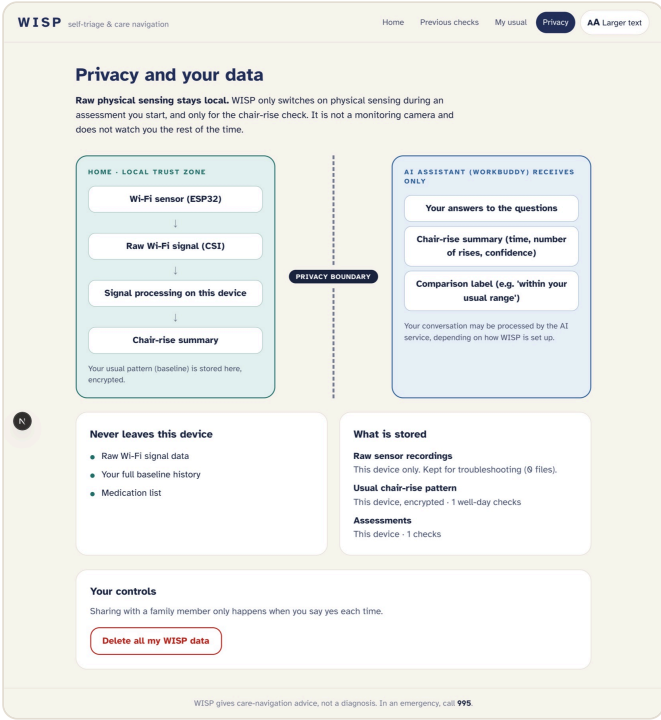
Caregiver summary. Exact message preview; shared only after an explicit “yes”. No sensor data.



Previous checks. Past outcomes, follow-ups, and whether the chair-rise check was used.



My usual chair-rise (baseline enrolment). Three well-day checks with the same chair and position.



Privacy. Home trust zone vs what the AI receives, what is stored, delete-my-data.

WISP

self-triage & care navigation

HomePrevious checksMy usualPrivacyAA Larger text

Developer / demo diagnostics

Not part of the patient experience. Raw CSI is only visualised here.

Sensor source

Provider: replay · mode **RECORDED REPLAY** · available

Switch with WISP_SENSOR_MODE=esp32

WISP_SERIAL_PORT=/dev/tty_ when starting the API. Replayed sessions are always labelled as recorded.

Next check plays: persona default (today) ▾

Queue "someone walks through" (Mdm Tan)

Queue interference (Mdm Siti)

Replay speed: 4x ▾

☐ Show live rise count (1/5...5/5) to the patient — only if validated as reliable

Agent

☒ Built-in agent — offline stand-in that calls the same tools (for rehearsal / no network).

☐ Tencent WorkBuddy via MCP — new assessments wait for WorkBuddy to attach with get_active_session. Conversation happens in WorkBuddy; this screen shows the check, trace and result.

Reset demo data

Measurements

id	session	ok	total s	per rise	conf	1-person	mode	recording
m_d8d1fa6d9d3a	enrol_lmr_lim	✓	10.06	1.05, 2.00, 2.00, 2.12, 1.93	1.00	1.00	synthetic_recorded	lin_baseline_1
m_857bfef441fb	s_c8b39f548638	✗	—	—	0.76	0.39	synthetic_recorded	tan_interference
multiple_people_detected								
m_59c36c9772b9	s_b64c54258628	✓	13.06	2.37, 2.76, 2.62, 2.78, 2.53	1.00	1.00	synthetic_recorded	siti_today
m_c1982a3d36df	s_7b9926148c06	✓	15.78	2.47, 2.9, 3.16, 3.64, 3.63	1.00	1.00	synthetic_recorded	tan_today
m_87528824356	enrol_mdms_siti	✓	12.78	2.31, 2.68, 2.68, 2.7, 2.41	1.00	1.00	synthetic_recorded	siti_baseline_3
m_f4a95c9a4fb9	enrol_mdms_siti	✓	13.2	2.47, 2.72, 2.72, 2.84, 2.45	1.00	1.00	synthetic_recorded	siti_baseline_2
m_4c58deea1b63	enrol_mdms_siti	✓	12.94	2.39, 2.7, 2.6, 2.8, 2.45	1.00	1.00	synthetic_recorded	siti_baseline_1
m_86ee85c4b122	enrol_mdms_tan	✓	11.26	2.07, 2.26, 2.34, 2.42, 2.17	1.00	1.00	synthetic_recorded	tan_baseline_3
m_477c94cb489	enrol_mdms_tan	✓	11.8	2.19, 2.46, 2.38, 2.58, 2.19	1.00	1.00	synthetic_recorded	tan_baseline_2
m_f38eb585580b	enrol_mdms_tan	✓	11.54	2.11, 2.4, 2.36, 2.4, 2.27	1.00	1.00	synthetic_recorded	tan_baseline_1

Motion energy (PCA of CSI amplitude) — shaded: detected test window

Pipeline features

pc_concentration	0.98
gait_band_ratio	0.682
trough_residual	0.24
cycle_shape_corr	0.912
first_cycle_ratio	0.842
pre_motion	0.039
stand_peaks	5
snr	3.26
rest_energy	0.206
subcarriers_used	52

Posture signal (distance from seated state) — dashed: detected standing phases

Record ground truth

Stopwatch or phone-video time for this session (seconds).

Save

Sensor validation (ground truth recorded in this deployment)

Only sessions with recorded ground truth are included. Synthetic replays are labelled and must not be reported as real accuracy.

Sessions	Failures	MAE (s)	Median abs err (s)	Mean confidence
0	0	—	—	—

Audit log (latest 60)

```
10:41:21 PM s_c8b39f548638 rule_engine care_tier_decided · decide_care_tier - ABSTAIN
10:41:21 PM s_c8b39f548638 local_agent tool_call · compare_to_baseline - measurement_unreliable
10:41:21 PM s_c8b39f548638 sensing measurement_complete · run_functional_assessment - multiple_people_detected
10:41:15 PM s_c8b39f548638 sensing sensing_active - replay
10:41:15 PM s_c8b39f548638 patient patient_ready_for_check
10:41:14 PM s_c8b39f548638 local_agent assessment_selected - 5x5TS
10:41:14 PM s_c8b39f548638 local_agent tool_call · run_functional_assessment - awaiting_patient
10:41:14 PM s_c8b39f548638 local_agent tool_call · check_assessment_eligibility - allowed
10:41:14 PM s_c8b39f548638 rule_engine red_flag_screen · screen_red_flags - passed
10:41:14 PM s_c8b39f548638 local_agent tool_call · record_case_facts - ok
10:41:14 PM s_c8b39f548638 rule_engine red_flag_screen · screen_red_flags - passed
10:41:14 PM s_c8b39f548638 local_agent tool_call · record_case_facts - ok
10:41:13 PM s_c8b39f548638 local_agent agent_decision · log_decision - Physical function check
10:41:13 PM s_c8b39f548638 local_agent tool_call · screen_red_flags - passed
10:41:13 PM s_c8b39f548638 rule_engine red_flag_screen · screen_red_flags - passed
10:41:13 PM s_c8b39f548638 local_agent tool_call · record_case_facts - ok
```

WISP gives care-navigation advice, not a diagnosis. In an emergency, call 995

WISP

HomePrevious checksMy usual

PrivacyAA Larger text

Assessment · Mdm Tan

RECORDED SENSOR SESSION · SYNTHETIC CSI

Physical sensing COMPLETE

RecommendationConversation

Same-day careConfidence: High

Please be seen today

See your regular GP or a polyclinic today.

When: Today

Find care

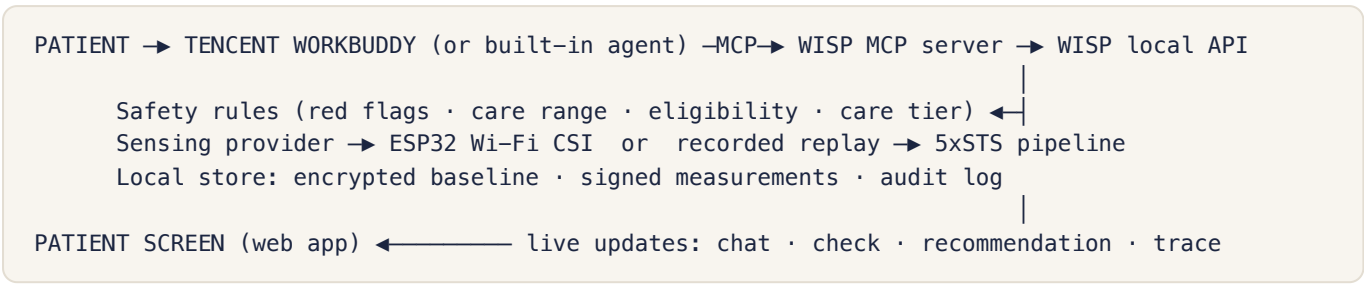
Phone layout. The recommendation at 390 px wide; the decision trace folds into an expandable panel.

Developer diagnostics (not shown to patients). Sensor source and replay controls, measurement table, motion-energy and posture charts with the detected test window and five standing phases, pipeline features, ground-truth entry, and the full audit log.

WISP — progress report

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4. How it is built



Who decides what

AI AGENT (WORKBUDDY)	FIXED SAFETY RULES (CODE)
Holds the conversation; turns words into structured facts; notices missing information; chooses whether a physical check is worth doing; explains the result; manages re-checks and sharing.	Emergency screening; whether sensing is allowed; care floor and ceiling; final care tier; abstention. The AI cannot skip, reorder or override these.

Safety guarantees (each covered by automated tests)

- **Sensing can raise urgency, never lower it** below the symptom-based floor.
- **Any warning sign** → **emergency**, and the sensor locks for the rest of the check; a previous normal result never overrides it.
- **No sensing before the warning-sign screen passes**, and only with a one-time permission tied to that check.
- **Home care only with a reliable, normal measurement** compared with a real baseline; unreliable readings are never used.
- **The agent cannot invent measurements**: they are produced by the sensor service, signed, and bound to the session.
- **Raw Wi-Fi data never leaves the device**; the agent only receives a summary, a comparison label and confidence.

5. Evaluation so far

All results below use synthetic data and small samples. No real-world sensing accuracy or population-level fairness is claimed.

EVALUATION	RESULT
End-to-end test conversations	15/15 right care level and right measure / don't-measure choice
Fairness: change only age, sex, living situation, language preference or phrasing (incl. Singlish, Malay- and Mandarin-mixed)	0 of 51 variants changed the result
Warning signs detected from free text (each sign is also asked directly)	14/15 detected, 0 false alarms; missed one Malay-only sentence
Chair-rise timing on synthetic Wi-Fi data (80 sessions)	Average error 0.21 s; 7 sessions wrongly rejected
Detecting a second person walking through (60 synthetic sessions)	30/60 rejected (50%) — known limitation of a single Wi-Fi link
Real participants with stopwatch / video timing	None yet — entry form and accuracy report are ready

6. Open items and next steps

- **Connect Tencent WorkBuddy** to the MCP server and run the scenarios with it (setup and agent instructions are ready).
- **Set up the two ESP32 boards**, record real chair-rise sessions with stopwatch timing, and re-tune the pipeline on real data.
- **Replace the synthetic demo recordings** with real captures before the judging demo.
- **Improve second-person detection** (e.g. a second receiver); meanwhile WISP asks whether anyone else is moving.
- **Clinical review** of the rule thresholds, which are prototype values.
- Arm use is currently self-reported; clinic availability is not live; caregiver delivery is simulated.

Run locally with `./scripts/demo.sh` (add `--mcp` for WorkBuddy) and open `http://localhost:3000`. Full details are in the repository README and docs/.